esprit Responsible Learning

HONORIS UNITED UNIVERSITIES

the rule of three

Level : 2A

Team C++

AY: 2020/2021

United Nations Educational, Scientific and Cultural Organization · UNESCO Chair "Project-based learning" ESPRIT School of engineering, Tunisia

EUR-ACE® Délivrée par la Commission des Titres d'Ingénieur

CONFÉRENCE DES GRANDES ÉCOLES

cdio CONCEIVE DESIGN IMPLEMENT OPERATE

Outline of the chapter

- Reminder: Collection of polymorphic objects
- Redefining the destructor
- Redefinition of the copy constructor
- Redefinition of the assignment operator
- To remember

Reminder: Collection of polymorphic objects

main.cpp

```
int main()
{
    Account C1(123, 1000);
    C1.display();
    cout<<"#################################################"<<endl;

    CurrentAccount CC1(234,2000);
    CC1.display();
    cout<<"#################################################"<<endl;

    SavingAccount CE;
    CE.display();
    cout<<"#################################################"<<endl;

    Bank B;

    cout<<"################ polymorphism ################"<<endl;
    B.add(CC1);
    B.add(C1);
    B.add(CE);

    cout<<"#################### Banque B ####################"<<endl;
    B.display();
```

Main and object B must not manipulate the same variables!

➔ **The variables of the main added in the bank B**

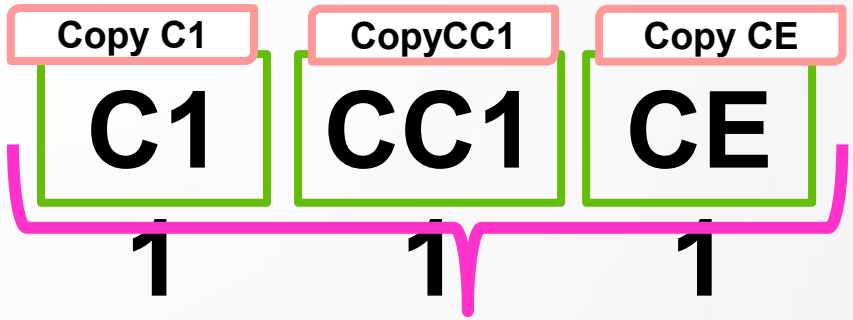
3

Reminder: Collection of polymorphic objects

main.cpp

```
int main()
{
    Account C1(123, 1000);
    C1.display();
    cout<<"##################################################"<<endl;

    CurrentAccount CC1(234,2000);
    CC1.display();
    cout<<"##################################################"<<endl;

    SavingAccount CE;
    CE.display();
    cout<<"##################################################"<<endl;

    Bank B;

    cout<<"################ polymorphism ###############"<<endl;
    B.add(CC1);
    B.add(C1);
    B.add(CE);

    cout<<"#################### Banque B ####################"<<endl;
    B.display();
```

```
bool Bank::add (SavingAccount const & C)
{
    if (searching(C.GetRIB())!=tab.end())
    {
        return false;
    }

    Account *p=new SavingAccount(C);
    tab.push_back(p);

    return true;

}
```



Add copies of C1, CC1 and CE (variables of the main) in object B (C11, CC11, CE1)

Redefinition of the destructor

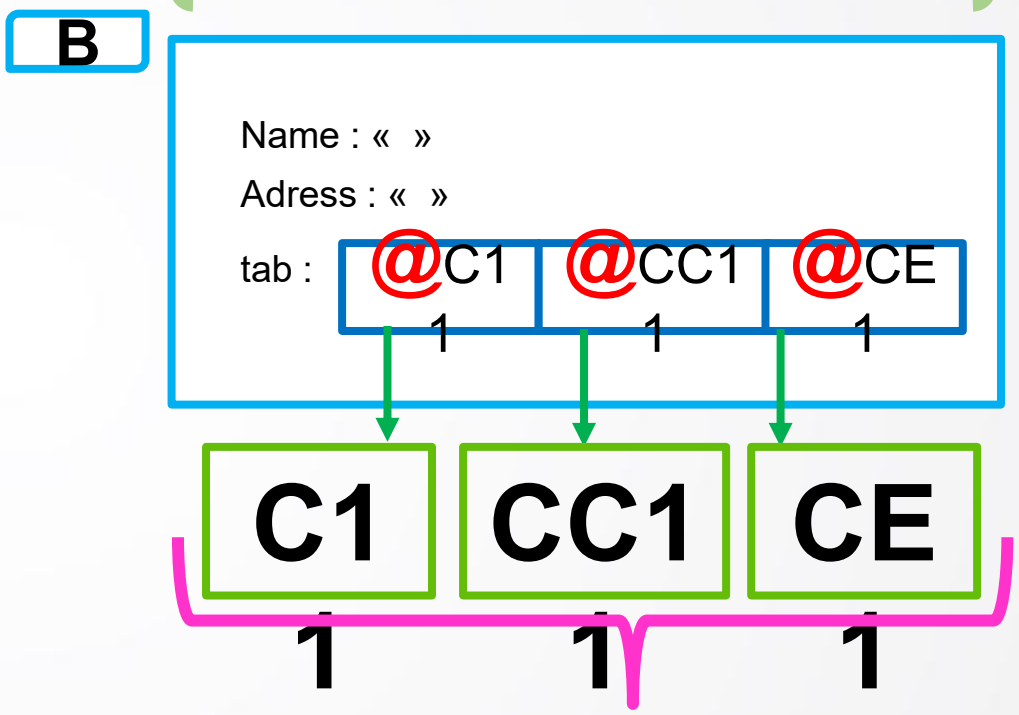
main.cpp

```
int main()
{
    Account C1(123, 1000);
    C1.display();
    cout<<"##################################################"<<endl;

    CurrentAccount CC1(234,2000);
    CC1.display();
    cout<<"##################################################"<<endl;

    SavingAccount CE;
    CE.display();
    cout<<"##################################################"<<endl;

    Bank B;

    cout<<"############## polymorphism ####
    B.add(CC1);
    B.add(C1);
    B.add(CE);


    ################## Banque B ##########
    B.display();
```

Static Allocation

Call to the **destructor** of the Bank class

Bank B; // Instantiation of the object B



These are copies of C1, CC1 and CE

Redefinition of the destructor

bank.h

```
~Bank();
```

bank.cpp

```
Bank::~Bank()
{
      cout<<"I am the destroyer of the Bank class"<<endl;
}
```

main.cpp

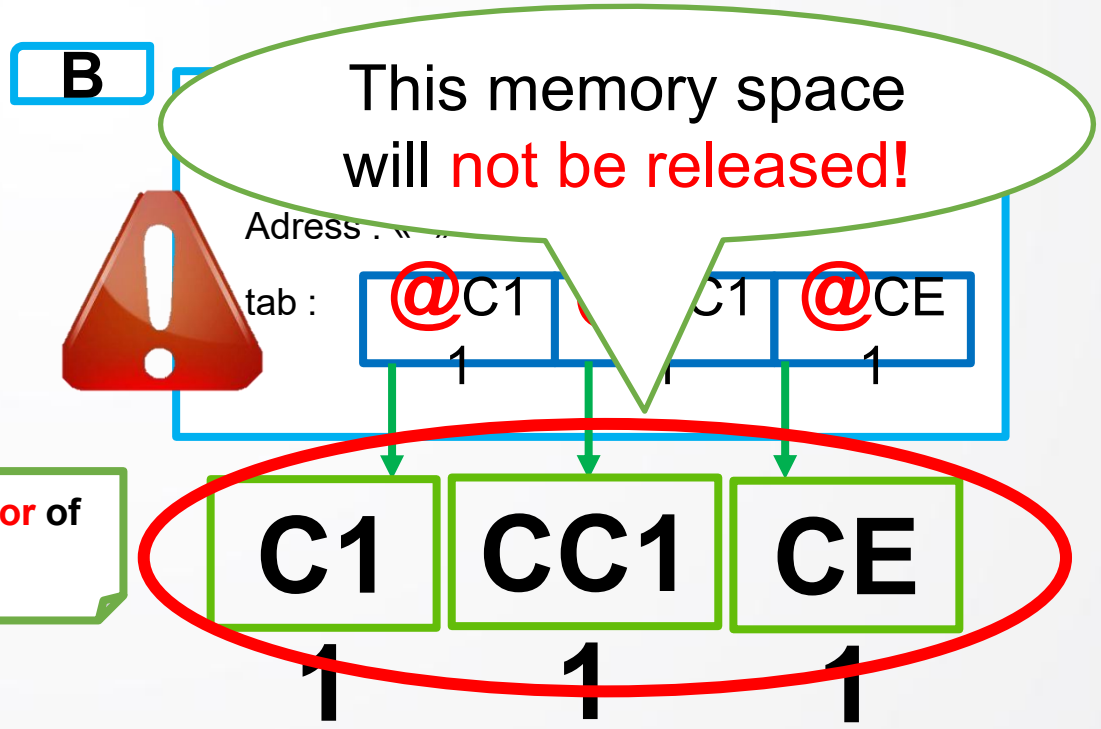
```
I am the destroyer of the Bank class
```

Redefinition of the destructor

```
Banque B;

B.add(C1);

B.add(CC1);

B.add(CE);

B.Display();

return 0;
}
```

Call to the **destructor** of the Bank class

- Call of the destructor of the class **String**
- Call of the destructor of the class **String**
- Call of the destructor of the class **Vector**



Free this space in the destructor of the Bank class

Redefinition of the destructor

bank.h

```
~Bank();
```

bank.cpp

```
Bank::~Bank()
{
    cout<<"I am the destroyer of the Bank class"<<endl;
}
```

```
Bank::~Bank()
{
    cout<<"I am the destroyer of the Bank class"<<endl;

    vector <Account*>::iterator i ;

    for(i=tab.begin(); i!=tab.end(); i++)
    {
        delete (*i);
    }
}
```

Redefinition of the destructor

bank.cpp

```
Bank::~Bank()
{
    cout<<"I am the destroyer of the Bank class"<<endl;

    vector <Account*>::iterator i ;
    for(i=tab.begin(); i!=tab.end(); i++)
    {
        delete (*i);
    }
}
```

main.cpp

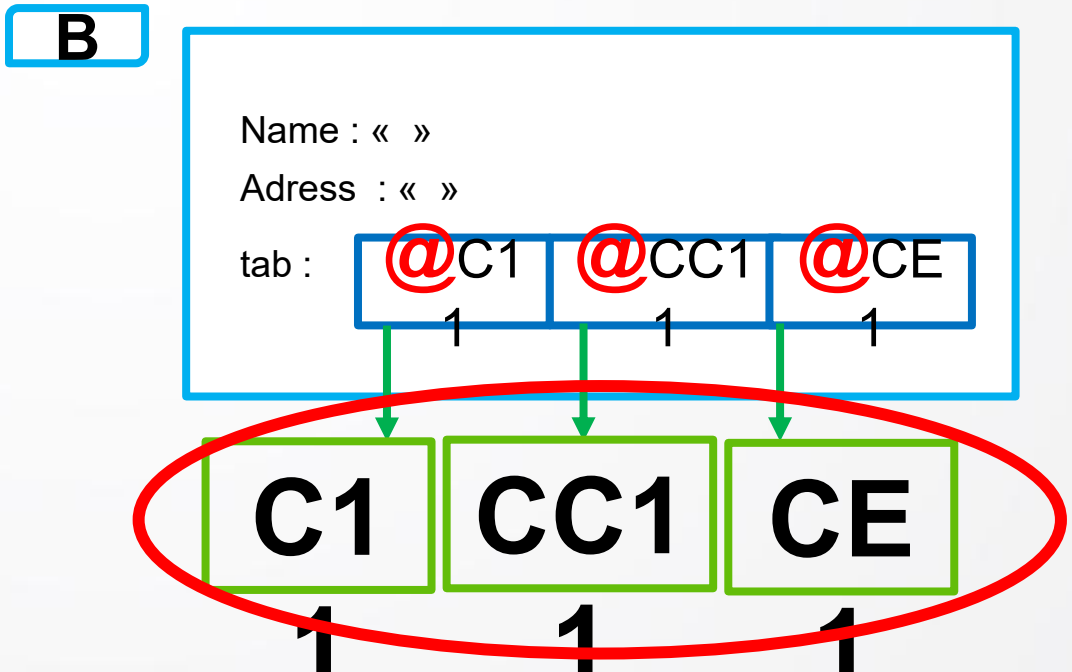
```
I am the destructor of the class Account
I am the destructor of the class Current account
I am the destructor of the class Account
I am the destructor of the class Current account
I am the destructor of the class Account
```

Redefinition of the destructor

```
Banque B;

B.add(C1);

B.add(CC1);

B.add(CE);

B.Display();

return 0;
}
```

Call of the **destructor** of the class Bank

- **delete (*i);**
- Call of the destructor of the class **String**
- Call of the destructor of the class **String**
- Call of the destructor of the class **Vector**



Redefinition of the copy constructor

main.cpp

```
cout<<"#################### Bank B ####################"<<endl;
B.Display();


cout<<"############ Constructor by copy ############"<<endl;

Banque A(B);

A.Display();
```

Call **to the copy constructor** of the Bank class

```
#################### Bank B ####################
RIB : 234
Pay : 1.29051e-306
RIB : 123
Pay : 1.29035e-306
RIB : 0
Pay : 5
rate : 0
############## Constructor of copy ############
RIB : 234
Pay : 1.29051e-306
RIB : 123
Pay : 1.29035e-306
RIB : 0
Pay : 5
rate : 0
```

Original

COPy

Redefinition of the copy constructor

main.cpp

```
Bank A(B);

B.modify(123);
cout<<"############ Bank B ############"<<endl;
B.Display();
cout<<"############ Bank A copy of Bank B ############"<<endl;
A.Display();
```

Modification of the content of Bank B

bank.cpp

```
bool Banque::modifier (long RIB)const

{
    vector <Compte*>::const_iterator i ;

    for(i=tab.begin(); i!=tab.end(); i++)
    {
        if((**i).GetRIB()== RIB)
        {
            (**i).Setsolde(55555);
            return true;
        }
    }
    return false;
}
```

Redefinition of the copy constructor

main.cpp

```
Bank A(B);

B.modify() ;
cout<<"############ Bank B ############"<<endl;
B.Display();
cout<<"############ Bank A copy of Bank B ############"<<endl;
A.Display();
```

Modification of the content of Bank B

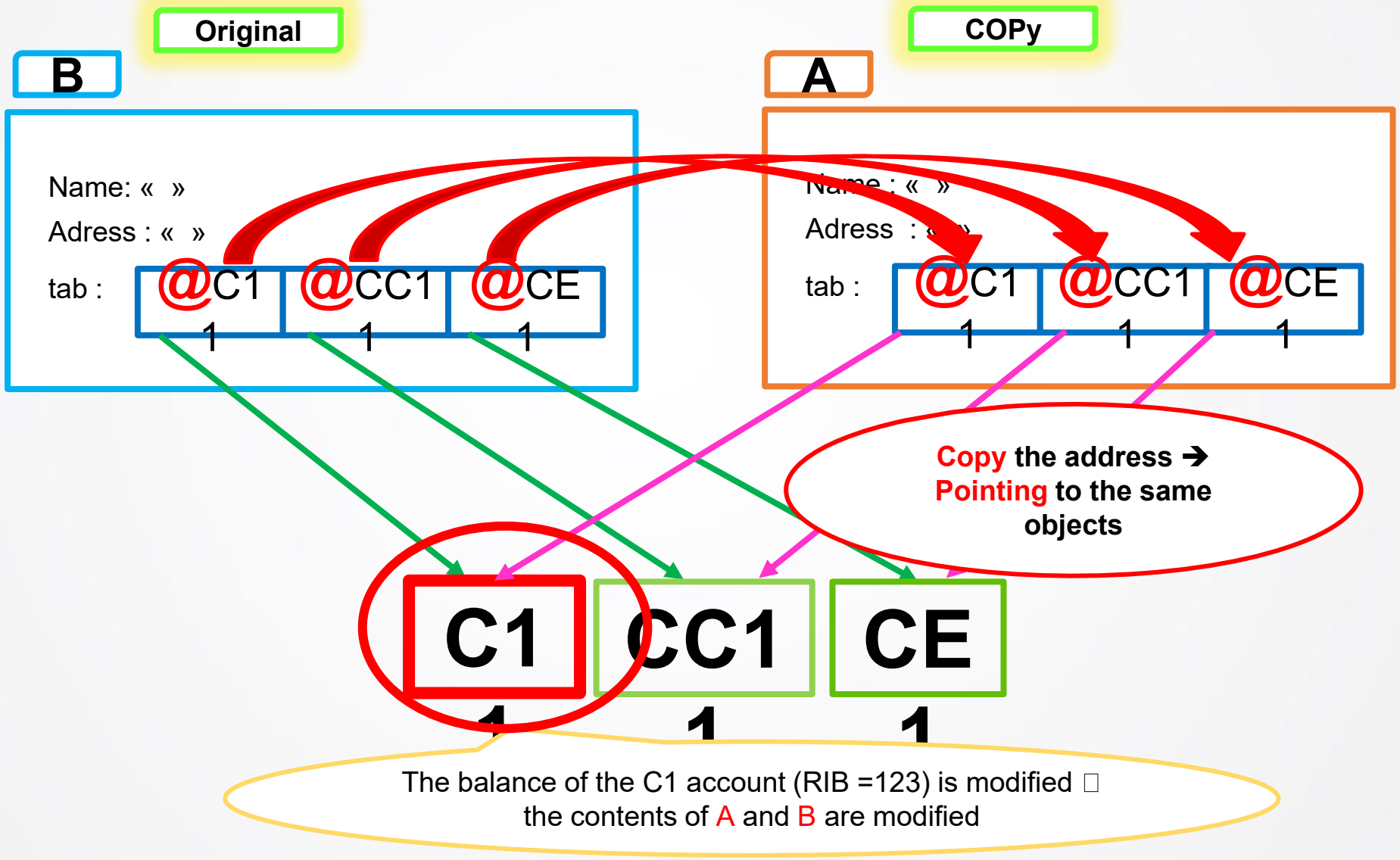
```
######################## Banque B ########################
RIB  : 123
Solde  : 55555
RIB  : 234
Solde  : 2000
Seuil  : 200
RIB  : 0
Solde  : 5
taux  : 0
############### La banque A copie de B ###############
RIB  : 123
Solde  : 55555
RIB  : 234
Solde  : 2000
Seuil  : 200
RIB  : 0
Solde  : 5
taux  : 0
```

B.modify();

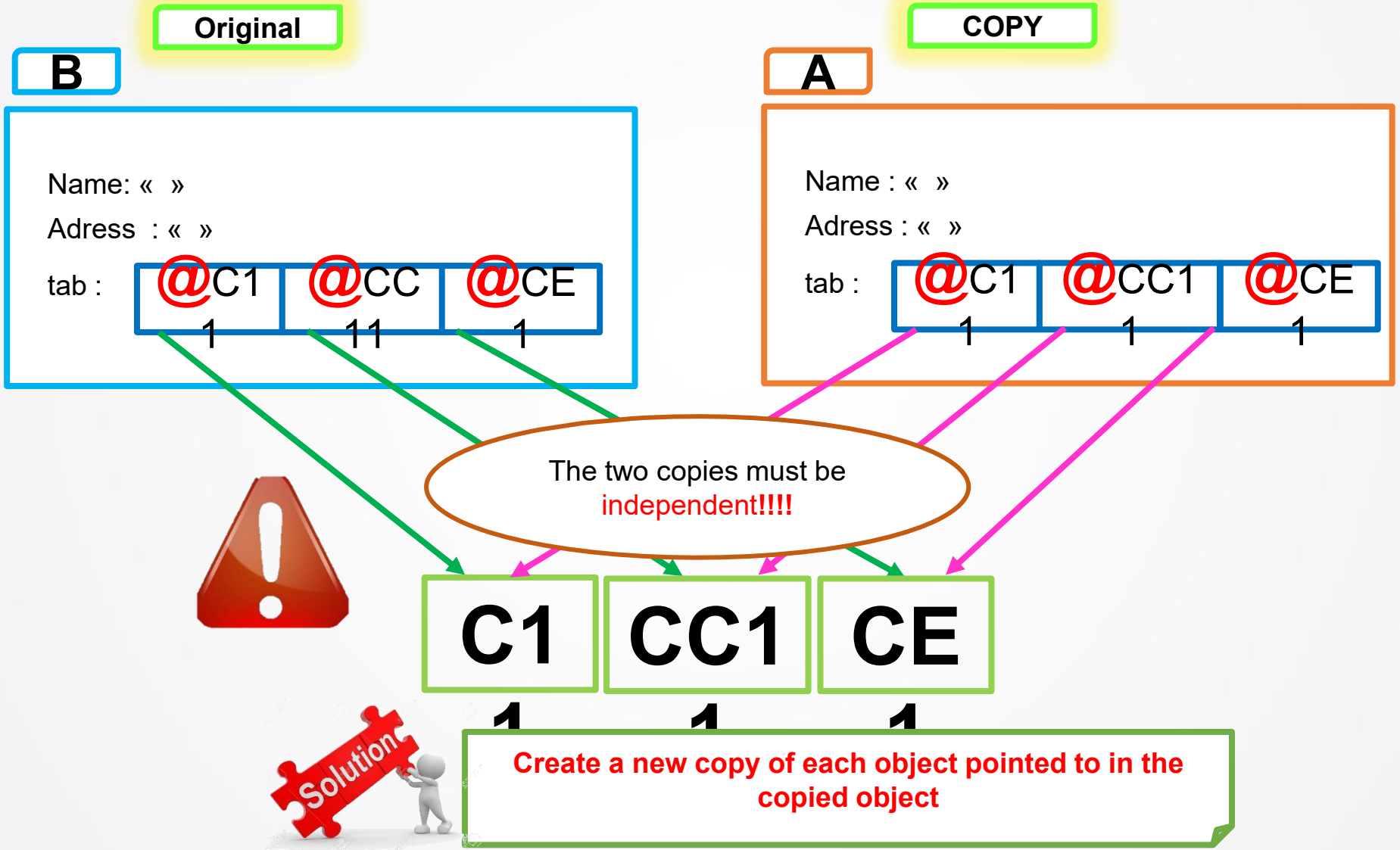
Original Modified

Copy Modified !!!

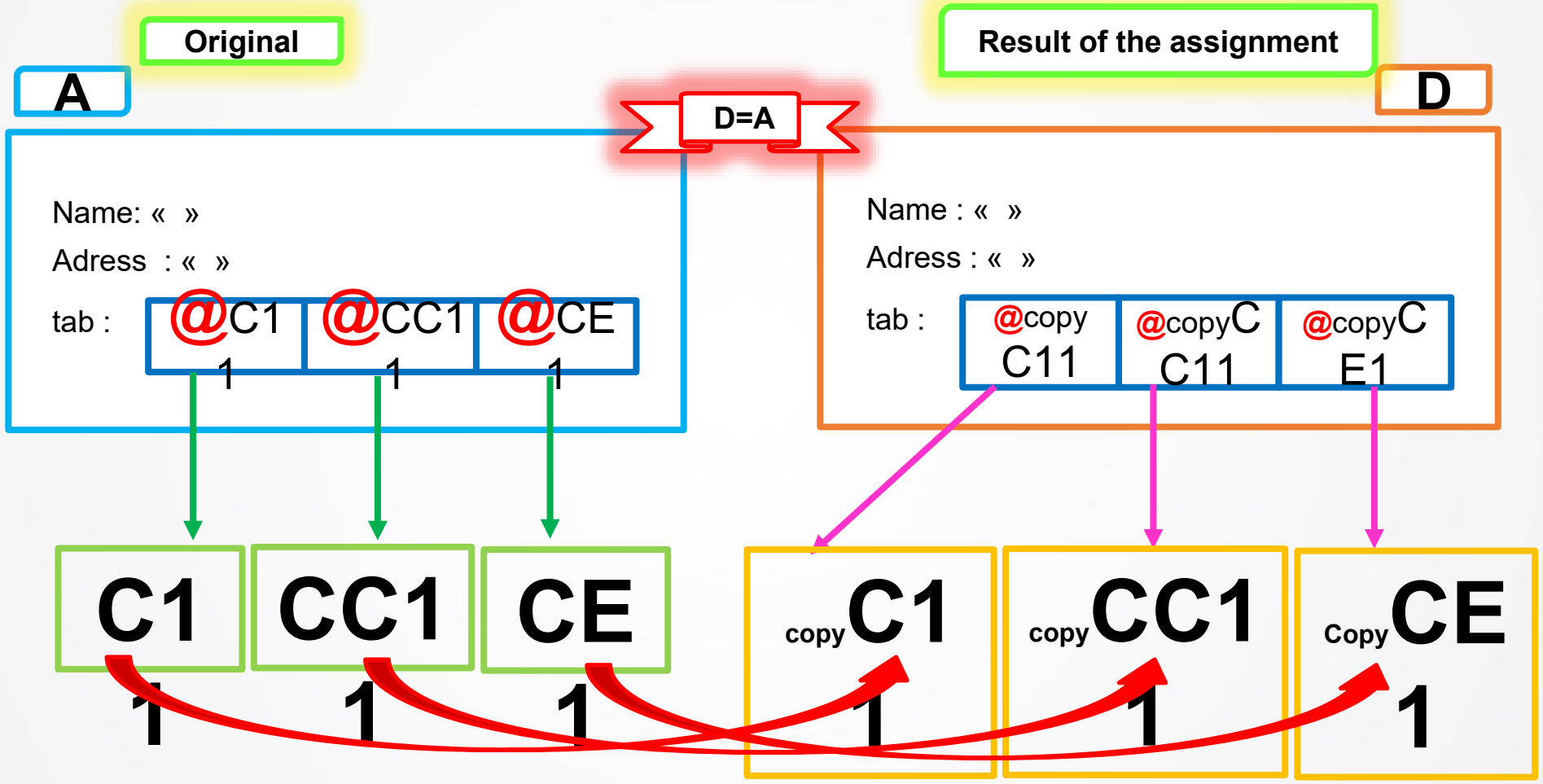
Redefinition of the copy constructor



Redefinition of the copy constructor



Redefinition of the assignment operator



Redefinition of the assignment operator

bank.h

```
Bank&operator=(const Bank&B)
```

bank.cpp

```
Bank::Bank(const Bank &B)
{
    Account *p;
    for(vector<Account *>::const_iterator i=B.tab.begin(); i != B.tab.end(); i++)
    {
        if(typeid(**i)==typeid(Account))
        {
            p=new Account(**i);
        }
        else if(typeid(**i)==typeid(savingsaccount))
        {
            p=new savingsaccount(static_cast<const savingsaccount&> (**i));
        }
        else
        {
            p=new currentaccount(static_cast<const currentaccount&>(**i));
        }
        tab.push_back(p);

    }
    return *this;
```

Creation of new copies

Constructor by copy

Expl : A = B =C ;

Redefinition of the assignment operator

main.cpp

```
A.modifier(234);
cout<<"#################### Banque A ####################"<<endl;
A.afficher();
cout<<"#################### Banque D ####################"<<endl;
D.afficher();
```

Modification of the content of Bank A

```
####################### Bank A #######################
RIB  :  234
Pay  :  55555
RIB  :  123
Pay  :  55555
RIB  :  0
Pay  :  5
rate  :  0
#################### Banque D ####################
RIB  :  234
Pay  :  1.29051e-306
RIB  :  123
Pay  :  55555
RIB  :  0
Pay  :  5
rate  :  0
```

Original

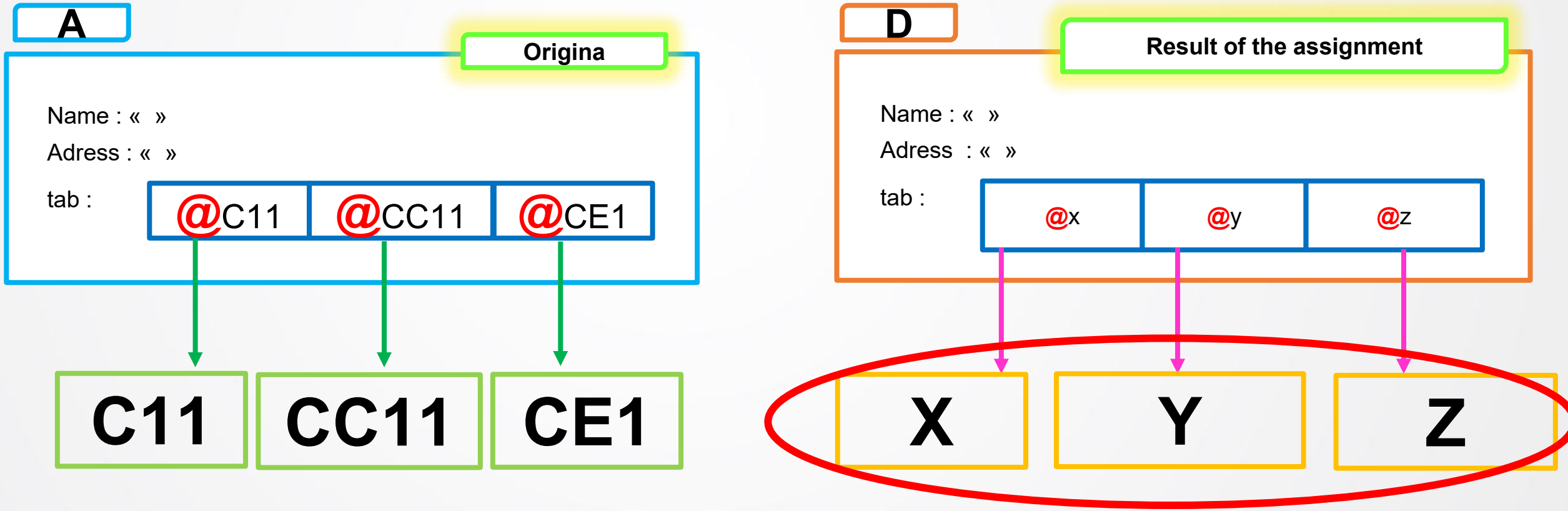
A.modifier();

Result of the assignment

Bank D contents NOT modified

Redefinition of the assignment operator

If D already references other objects … !!



Redefinition of the assignment operator

main.cpp

```
cout<<"################ Assignment A in D  ################"<<endl;
Bank D;

Account X;
currentaccount Y (456,33);
currentaccount Z (22222,2000);

D.add(X);
D.add(Y);
D.add(Z);

cout<<"#################### bank D ####################"<<endl;

D.display();

D=A;//D.operator=(A);

A.change(234);

cout<<"#################### Bank A ####################"<<endl;
A.display();
cout<<"#################### Banque D ####################"<<endl;
D.display();
```

D contains copies of objects **X, Y et Z**

assign **A** in **D**

Redefinition of the assignment operator

```
################### bank D ###
RIB : 0
Pay : 0
RIB : 456
Pay : 5.16106e-305
RIB : 22222
Pay : 6.28154e-317
```

D contains copies of objects
X, Y et Z

```
Bank D;

Account X;
savingsaccount y;
y.enter();
currentaccount Z (22222,2000,200);

D.add(X);
D.add(Y);
D.add(Z);

D.display();
```

```
######################## Bank A ##########
RIB  :  234
Pay  :  55555
RIB  :  123
Pay  :  55555
RIB  :  0
Pay  :  5
rate  :  0
```

Original

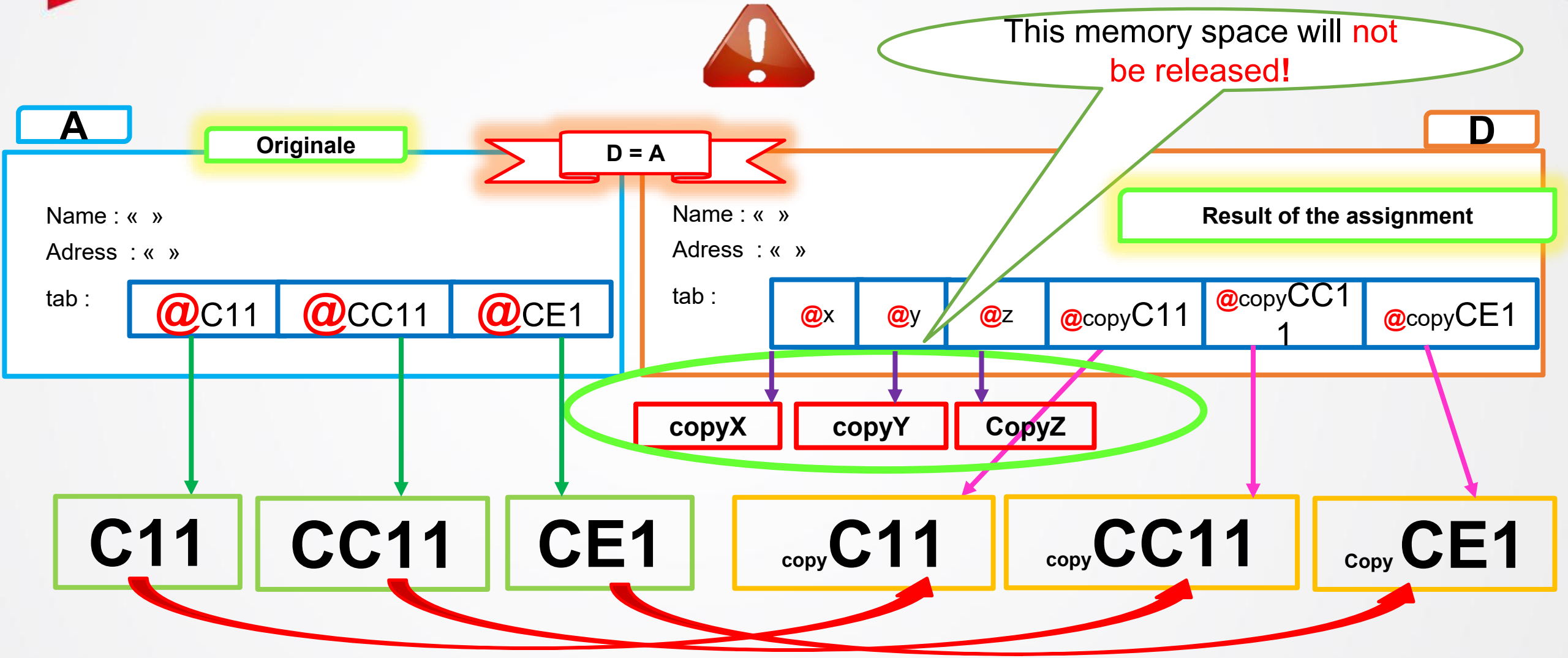
Result of the assignment

```
D=A;  //D.operator=(A);

A.change(234);
A.display();
D.display();
```

```
RIB  :  234
Pay  :  1.29051e-306
RIB  :  123
Pay  :  55555
RIB  :  0
Pay  :  5
rate  :  0
```

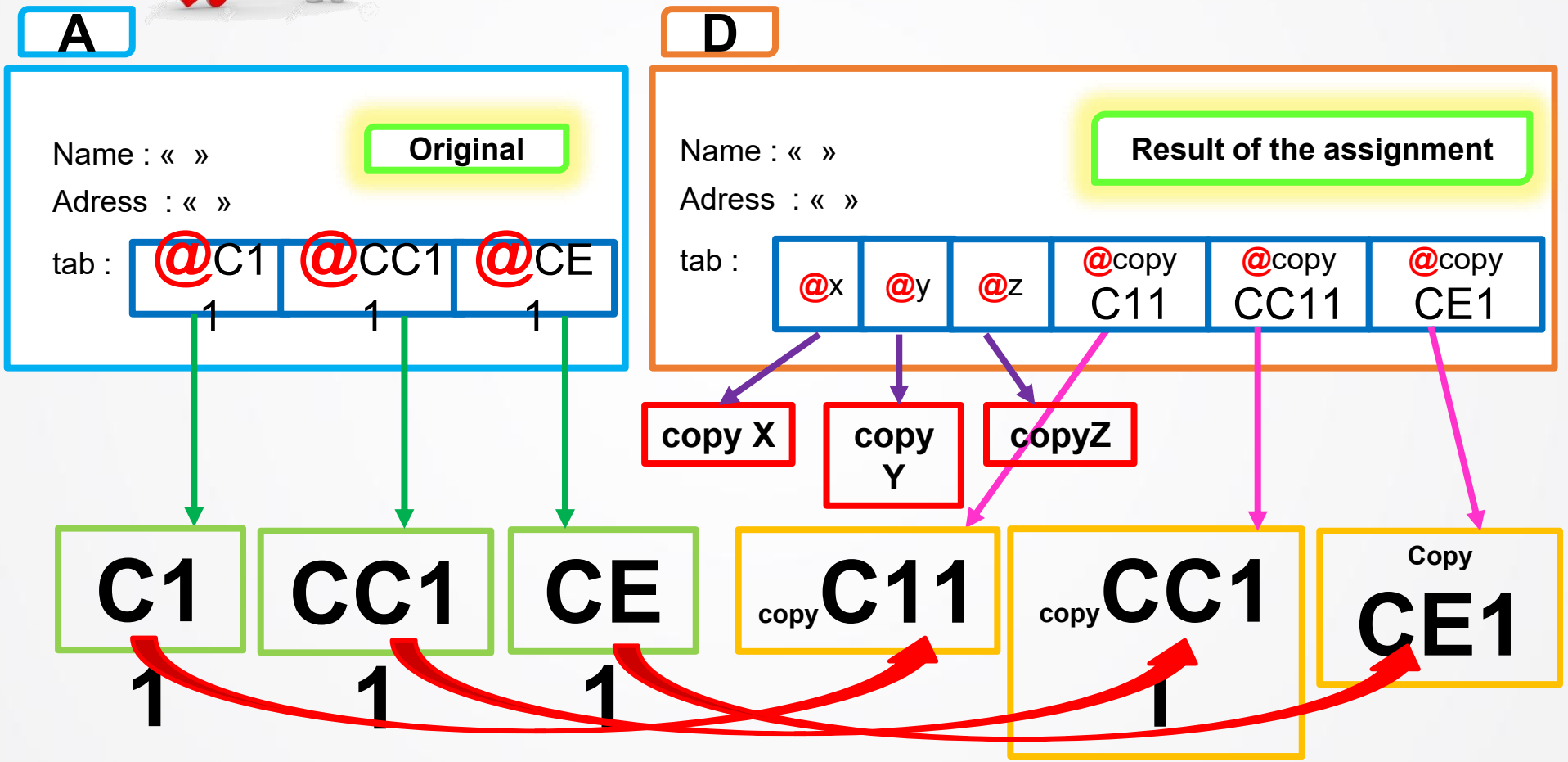
Result of the assignment **A** in **D**
D contains copies of
X, Y , Z C11, CC11, CE1 !!!

Redefinition of the assignment operator



Redefinition of the assignment operator

Delete the old content of D before making the assignment



Redefinition of the assignment operator

bank.cpp

```
Bank& Bank::operator=(const Bank&B)
{
   if(&B!=this)
      {
          vector <Account*>::iterator i ;
          for(i=tab.begin(); i!=tab.end(); i++)
          {
               delete (*i);
          }
          tab.clear();

          Account*p;
          for(vector<Account*>::const_iterator i=B.tab.begin(); i != B.tab.end(); i++)
          {
               if(typeid(**i)==typeid(Account))
               {
                    p=new Account(**i);
               }
               else if(typeid(**i)==typeid(currentaccount))
               {
                    p=new currentaccount(static_cast<const currentaccount&>(**i));
               }
               else
                    p=new savingsaccount(static_cast<const savingsaccount&>(**i));
               tab.push_back(p);
          }
      }
      return *this;
}
```

Destructor

Redefinition of the assignment operator

bank.cpp

```
Bank& Bank::operator=(const Bank&B)
{
  if(&B!=this)
    {
        vector <Account*>::iterator i ;
        for(i=tab.begin(); i!=tab.end(); i++)
        {
            delete (*i);
        }
        tab.clear();

        Account*p;
        for(vector<Account*>::const_iterator i=B.tab.begin(); i != B.tab.end(); i++)
        {
            if(typeid(**i)==typeid(Account))
            {
                p=new Account(**i);
            }
            else if(typeid(**i)==typeid(currentaccount))
            {
                p=new currentaccount(static_cast<const currentaccount&>(**i));
            }
            else
                p=new savingsaccount(static_cast<const savingsaccount&>(**i));
            tab.push_back(p);
        }
    return *this;
}
```

Avoid assigning an object to itself
Expl : A = A !!!

Destructor

Constructor of copy

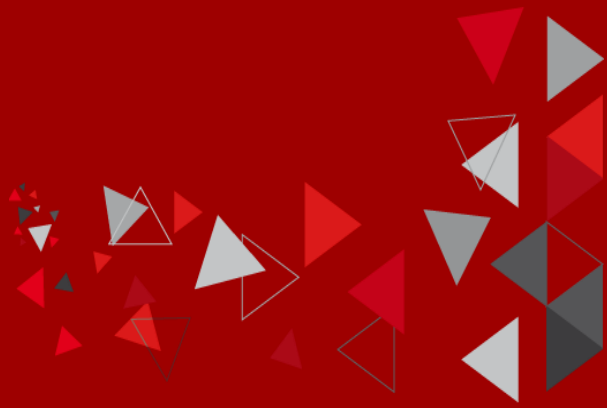
Expl : A = B =C ;

To remember

- « Defining **a copy constructor** for a class usually means that **the copy constructor, destructor and assignment operator** provided by default by the compiler **are not suitable for this class**. Therefore, all three methods will have to be systematically redefined as soon as one of them is. » *Course of C/C++, Christian Casteyde*

- « This rule, called **the rule of three**, will allow you to avoid bugs easily. » *Course of C/C++, Christian Casteyde*

- « This will be the case when some of the data of the objects have been **dynamically allocated**. A brutal copy of the fields of one object into another would only copy the pointers, **not the pointed data**. Thus, changing the data for one object would change the data for the other object, which would probably not be the desired effect. » *Course of C/C++, Christian Casteyde*



THANK YOU

Any Questions ?